

5 MW Containerized Data Center — Indonesia

Power & cooling infrastructure scope. 140 kW/rack, hybrid direct-liquid + air cooling. Feasibility review against US-basis financial model. Prepared 15 September 2026.

5,040 kW IT CAPACITY	36 RACKS @ 140 KW	1.18 ANNUAL PUE	\$31.1M CAPEX, POWER + COOLING
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Scope definition

One block = 5 MW. This document covers **power and cooling infrastructure only**: electrical distribution, UPS, generation, heat rejection, liquid cooling plant, containers, and supporting civil works.

Excluded: IT hardware (servers, GPUs, storage), network switching and optics, software, and operating staff costs.

1. Financial Model Review — Indonesia Applicability

1.1 What the source model implies per 5 MW block

Metric	Per block	Derived value
Capital required	\$28,750,000	\$5,750 per kW
Steady-state revenue	\$11,040,000/yr	\$184/kW/yr (~\$0.184/W/month)
Net operating cash	\$7,877,500/yr	71.4% operating margin
Pre-tax IRR	20%	—
After-tax IRR	17%	—
Year-one deduction (3 blocks)	\$80,250,000	93% of \$86.25M capex
Year-one tax value (3 blocks)	\$29,692,500	Implied tax rate ~37%

Core finding

A 93% year-one deduction is **US Section 168(k) bonus depreciation**. The ~37% implied rate is US federal plus state. Neither exists in Indonesia. The 17% after-tax IRR is substantially a tax-shield result, and that shield does not transfer.

1.2 Assumption-by-assumption transfer test

Assumption	US basis	Indonesia reality	Effect
93% year-one depreciation	Sec. 168(k) bonus	Not available. Buildings 20-yr straight-line; equipment Group 2 (8-yr) / Group 3 (16-yr) declining balance	Negative — removes the principal driver of after-tax IRR
~37% tax rate	Federal + state	22% corporate income tax	Mixed — lower shield value, lower tax on income
Power cost	~\$0.070–0.090/kWh industrial	PLN B3/I3 approx. IDR 1,035–1,115/kWh $\approx$ \$0.065–0.070/kWh at IDR 16,300/USD	Positive
Grid availability	Interconnect measured in months	5 MW requires PLN 20 kV medium-voltage service. Java and Batam have headroom; outer islands do not	Schedule risk, not cost risk
Land and civil	US market rates	Materially cheaper; largely irrelevant under a container deployment	Positive, small
Labour	~\$150k/engineer	IDR 15–30 million/month for DC engineering staff	Positive
Equipment import	Not applicable	VAT 11–12%, import income tax (PPh 22) 2.5%, duty 0–5%. Waivable in a bonded zone or under KITE	Adds 5–12% to imported plant unless bonded
Revenue \$184/kW/yr	US colocation	Indonesian retail colocation approx. IDR 4–8 million/kW/month = \$245–490/kW/yr. AI hosting prices above this	Model revenue is conservative
Currency	USD throughout	Capex largely USD-denominated; revenue likely IDR unless contracts are written in USD	Genuine unhedged exposure
Regulatory	—	Gov. Reg. 71/2019 and Personal Data Protection Law 27/2022 drive localization for public-sector and parts of financial data	Positive — mandated domestic demand

1.3 Verdict

The project works in Indonesia, but for a different reason

US version is tax-driven. The 93% year-one deduction is the deal.

Indonesia version is revenue-driven. Cheap power, cheap labour, a data-localization mandate, an undersupplied market, and colocation pricing 1.3–2.6× the modelled rate.

Required model changes:

- Remove bonus depreciation. Apply Group 3 (16-yr) to building shell and Group 2 (8-yr declining) to mechanical and electrical plant.
- Apply for **Tax Holiday** through OSS/BKPM. Data centers fall within the pioneer-industry list, and capital expenditure above IDR 500 billion qualifies — a three-block build at \$86M clears this comfortably. A 5–20 year CIT exemption becomes the replacement for the US depreciation advantage.
- Site inside a bonded zone or Special Economic Zone to waive import duty and VAT on plant and to ease foreign ownership.
- Hedge foreign exchange, or write tenant contracts in USD.

Without a Tax Holiday, after-tax IRR falls roughly 3 percentage points below the 17% modelled and becomes materially back-loaded.

1.4 Site ranking

Rank	Location	Case for	Case against
1	Batam — Nongsa Digital Park	SEZ status with tax and import incentives. 1–3 ms to Singapore via submarine cable. Reliable PLN (b'right) supply. Absorbs demand displaced by Singapore's data center moratorium.	Smaller local market; depends on regional and international tenants
2	Jakarta / Cikarang / Karawang	Primary demand centre. IIX/OpenIXP and MIX peering on the doorstep.	Expensive land, congested Java-Bali grid in places, flood exposure
3	Kendal / Batang SEZ	Low-cost power on the Java north coast. New industrial estates with available land.	Weak connectivity and limited peering

Rule out any site requiring new PLN transmission construction. That adds 18–36 months to the programme.

2. Design Basis

Parameter	Value	Note
IT load	5,000 kW	One block
Rack density	140 kW	—
Rack count	36	5,040 kW installed
Liquid-cooled fraction	85%	119 kW/rack captured at the cold plate
Air-cooled fraction	15%	21 kW/rack — NICs, PSUs, DIMMs, storage
Direct-liquid heat load	4,284 kW	—
Air heat load	756 kW	—
Design ambient	35°C DB / 28°C WB	0.4% design condition, Jakarta / Batam
Target PUE	1.15 – 1.25	Annualised 1.18 estimated
Resilience	N+1, concurrently maintainable	Tier III equivalent

Why 85/15 and not 100% liquid

NVIDIA GB200 NVL72 and GB300 reference designs capture roughly 85–90% of heat at the cold plate. The balance leaves via air. Designing for full liquid capture strands the air-side capacity and leaves network interface cards and power supplies without adequate cooling.

3. Container Configuration

Total **18 containers** plus four outdoor equipment yards.

3.1 IT containers — 6 units

Item	Specification
Type	40 ft high-cube ISO, ruggedised, 12.19 × 2.44 × 2.90 m
Racks per container	6 (840 kW IT per container)
Rack	48U OCP ORV3 or 19" 750 × 1200 mm, 1,200 kg static rating
Layout	Single contained cold aisle, three racks each side
Liquid interface	Rear-door manifold with in-rack quick-disconnect, blind-mate (Stäubli or CPC)
Air handling	2 × 150 kW in-row CRAH, N+1 against a 126 kW air load
Loaded weight	Approx. 14,000 kg — requires an engineered pad, not pier blocks
Envelope	100 mm polyurethane, R-4.0 minimum, continuous vapour barrier (ambient RH exceeds 80%)
Fire	Aspirating smoke detection plus FM-200 or Novec 1230, cross-zoned per container
Leak detection	Rope sensors beneath the floor, per-rack drip tray with float switch, hard-interlocked to CDU isolation

3.2 Direct-to-chip loop (secondary)

Parameter		Value
Supply temperature		32°C
Return temperature		45°C
Delta-T		13 K
System mass flow		78.8 kg/s
System volume flow		approx. 4,730 L/min
Flow per container		approx. 790 L/min
Fluid	25% propylene glycol / 75% deionised water, conductivity below 1 µS/cm, with biocide and corrosion inhibitor	
Working pressure	3 bar secondary maximum, 6 bar primary	
Manifold	2 × DN65 flow and return per rack row, dual-path redundant loop	

Dew-point interlock — mandatory in this climate

At 28°C wet bulb and 80% relative humidity, dew point reaches 25–27°C. A 32°C secondary supply sits safely above it. The control loop must never reset secondary supply below dew point plus 2 K. Enforce this as a hard floor of 29°C **in CDU firmware**, not only in the building management system.

3.3 CDU containers — 2 units

Item	Specification
Type	20 ft ISO
CDU per container	3 × 1,500 kW liquid-to-liquid
Installed capacity	9,000 kW against a 4,284 kW load — 2N at CDU level
Approach temperature	3 K (primary 29°C in, secondary 32°C out)
Pumps	2 × 100% per CDU, variable speed, hot-swappable
Heat exchanger	Brazed plate, 316L stainless; titanium where condenser water is saline-adjacent
Filtration	50 µm bag plus 5 µm cartridge on secondary, side-stream
Control	Local PLC per CDU, Modbus TCP to BMS, dew-point-aware setpoint reset

4. Heat Rejection

4.1 Primary option — adiabatic dry cooler (recommended)

Item	Specification
Units	8 × 750 kW adiabatic V-bank dry cooler, N+1 against 5,290 kW rejection
Primary loop	29°C supply / 41°C return
Dry mode	Ambient below 26°C — nights and rainfall. Zero water consumption
Adiabatic mode	Ambient above 26°C — pre-cools entering air toward wet-bulb
Water consumption	15–25 m ³ /day in adiabatic mode. WUE approx. 0.15 L/kWh
Water treatment	Reverse osmosis or softened supply. Indonesian municipal water will scale the pads untreated
Fans	EC, 0–100% variable, below 65 dBA at 10 m
Footprint	Approx. 450 m ²

4.2 Alternative — open cooling tower with chiller backup

Delivers a better PUE (approximately 1.12) but consumes 8–12× the water, near 200 m³/day. It carries Legionella management obligations, and water permitting is difficult: Jakarta restricts groundwater extraction because of land subsidence, and municipal supply is not designed for this duty. **Viable only in Batam or an SEZ holding an industrial water contract.**

4.3 Chilled water for the air load

- 2 × 500 kW air-cooled screw chillers, N+1, on R-513A or R-1234ze. Avoid R-410A given HFC phase-down.
- 10°C supply / 16°C return to the in-row CRAH units.
- **Cross-connected to the CDU primary loop via three-way valve.** This provides trim capacity on days exceeding design ambient and is the safety valve for the liquid plant.

4.4 Thermal budget

Heat stream	Load	Rejection path
Direct liquid (chip)	4,284 kW	CDU → primary loop → adiabatic dry cooler
Air (in-rack residual)	756 kW	In-row CRAH → chilled water → air-cooled chiller
UPS and transformer losses	approx. 250 kW	Chiller and ambient
Total rejection	approx. 5,290 kW	8 × 750 kW dry cooler (N+1) + 2 × 500 kW chiller

5. Electrical

Item	Specification
Utility supply	PLN 20 kV medium voltage, dual feeder from separate substations where available
Transformers	2 × 4,000 kVA, 20 kV / 400 V, cast resin, Dyn11, outdoor containerised
Total design load	5,000 kW IT + approx. 1,100 kW mechanical = 6,100 kW → 7,625 kVA at 0.8 PF
UPS	4 × 1,500 kVA modular lithium-ion, N+1, 5 minutes autonomy
Battery chemistry	Lithium-ion. One-third the footprint, 10-year life, tolerates 30°C ambient. VRLA degrades rapidly at Indonesian temperatures
Generators	3 × 2,500 kVA prime-rated diesel, N+1, 48 hours on-site fuel
Distribution	Overhead busway, 800 A per IT container, dual A/B feed
Rack PDU	2 × 125 A 415 V three-phase (2 × 89 kW), metered with outlet-level switching. A 63 A PDU delivers only 45 kW and will not serve a 140 kW rack
Rack input	Busbar-fed. 48 V DC where OCP racks are used — better efficiency and fewer power supplies
Earthing	TN-S, below 1 ohm, surge protection Type 1 + 2. Ring earth plus four rods minimum per container

Two Indonesia-specific electrical risks

Lightning. The Bogor and Depok corridor records among the highest strike densities on Earth. Surge protection and earthing are primary design items, not an afterthought.

Generator run hours. PLN reliability varies widely by region. Assume more generator operation than a US model would predict, and budget fuel and maintenance accordingly. Commercial data centers must use non-subsidised diesel (Pertamina Dex); subsidised fuel is not lawfully available for this use.

6. Site and Civil

Item	Specification
Land area	Approx. 5,500 m ² (0.55 ha) including setbacks, generator yard, fuel storage and security perimeter
Pad	Reinforced concrete, 250 mm, 20 t/m ² bearing, graded 1:100 to drainage
Flood protection	Pad raised 1.2 m above the 50-year flood level. Non-negotiable on Java — the 2020 and 2026 flood events took facilities offline
Seismic	Per SNI 1726:2019. Seismic snubbers on all containers, flexible connections on all pipework
Stacking	IT containers single level only. Stack generator and utility containers if land is constrained
Security	2.4 m fence with razor topping, ANPR vehicle gate, mantrap at the meet-me room, 32-camera CCTV with 90-day retention
Site fire	Hydrant ring with jockey pump per SNI 03-1745. Foam system at the fuel yard
Permits	PBG (formerly IMB), SLF, UKL-UPL or AMDAL, environmental permit, PLN SLO certification, Komdigi PSE registration

Fibre entry and the meet-me room are provisioned within this scope as containerised space with dual diverse pathways. Active network equipment is a tenant or separate-package item.

7. PUE Estimate — Batam, Annualised

Component	Power (kW)
IT load	5,000
Dry cooler fans and primary pumps	210
CDU secondary pumps	95
Chiller (air load)	240
CRAH fans	60
UPS losses (96.5% eco mode)	175
Transformer and distribution losses	110
Lighting, auxiliary, security	30
Total facility power	5,920
PUE	1.184

Worst-case hour at 35°C dry bulb: approximately 1.28. Expected annualised range 1.18–1.20.

8. Capital Expenditure — Power and Cooling Only

Line item	USD
Land (0.55 ha, SEZ lease basis)	600,000
Civil — pad, drainage, flood works, fencing	1,800,000
6 × IT containers — shell, fit-out, racks, in-row CRAH	5,400,000
2 × CDU containers with 6 × 1,500 kW CDU	3,200,000
Heat rejection — dry coolers, chillers, primary pipework	3,600,000
Electrical — transformers, UPS, generators, switchgear, busway	9,800,000
Meet-me room containers, fibre entry, structured cabling	800,000
Fire suppression, security, BMS and DCIM	1,300,000
PLN connection and SLO certification	900,000
Design, project management, commissioning, permitting	2,200,000
Contingency at 10%	2,960,000
Total — power and cooling infrastructure	32,560,000
Cost per kW	6,512

Variance against the model

The source model assumes \$5,750/kW. This estimate lands at **\$6,512/kW — 13% higher**, a gap of approximately \$3.8 million per block.

Drivers: 140 kW/rack is high density and carries a liquid-infrastructure premium; import duty and VAT apply unless the site is bonded; flood-resilient civil works are mandatory on Java; 48-hour fuel storage exceeds typical US practice.

Mitigation: bonded-zone or SEZ status removes import duty and VAT on plant, bringing the figure to approximately **\$6,050/kW** — within 5% of the model.

9. Programme

Month	Activity
0 – 8	Site acquisition, PLN capacity application, permitting. PLN is the critical path — begin on day one
6 – 10	Civil works: pad, drainage, perimeter, fuel containment
4 – 12	Container fabrication, off-site and in parallel
10 – 13	Generator, transformer and switchgear delivery and energisation
13 – 15	Container placement and interconnection
15 – 16	Fluid fill, flush, passivation, pressure test at 1.5× working pressure
16 – 18	Commissioning L1–L5, integrated system test at 100% load bank, N-1 and thermal runaway scenarios
18+	Handover for IT deployment

Total approximately 18 months. Compressible to 12–14 months where PLN capacity already exists at the site — the principal argument for Batam and Nongsa Digital Park.

10. Items to Resolve Before Commitment

#	Item	Why it matters
1	Written PLN capacity confirmation at each candidate site	Determines whether the programme is 12 or 30 months. Secure before committing to land
2	Tax Holiday eligibility assessment via OSS/BKPM	Replaces the US depreciation advantage. Without it, after-tax IRR falls roughly 3 points
3	Bonded zone or SEZ status	Worth approximately \$460/kW in avoided import duty and VAT
4	Revenue currency and hedging policy	USD capex against IDR revenue is real exposure. IDR moved more than 20% in both 2013 and 2018
5	Industrial water contract for adiabatic cooling	Not municipal supply, not groundwater. Determines whether the recommended cooling option is available
6	Anchor tenant	5 MW fully speculative is aggressive for this market. A single tenant at 2–3 MW de-risks the block

Summary position

Building this in Indonesia is sound. The capital cost runs roughly 13% above the model, reducible to within 5% through SEZ siting. The tax structure must be rebuilt from the ground up — the US depreciation shield is unavailable, and a Tax Holiday must take its place.

The compensating advantage is that Indonesian colocation pricing runs 1.3–2.6× the revenue assumed in the model, against power at or below US industrial rates and a regulatory regime that mandates domestic capacity. The return case is intact; its source moves from the tax line to the revenue line.

Prepared 15 September 2026. Scope: power and cooling infrastructure for one 5 MW block. Excludes IT hardware, network equipment, software and operating staff. Costs are order-of-magnitude estimates in USD at IDR 16,300/USD and require vendor quotation before use in a committed budget. PLN tariffs, tax incentives and import treatment are subject to change and require confirmation with current regulation at the time of application.